

OPERATOR'S MANUAL

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CHAPTER THREE

CONTROLS AND INDICATORS

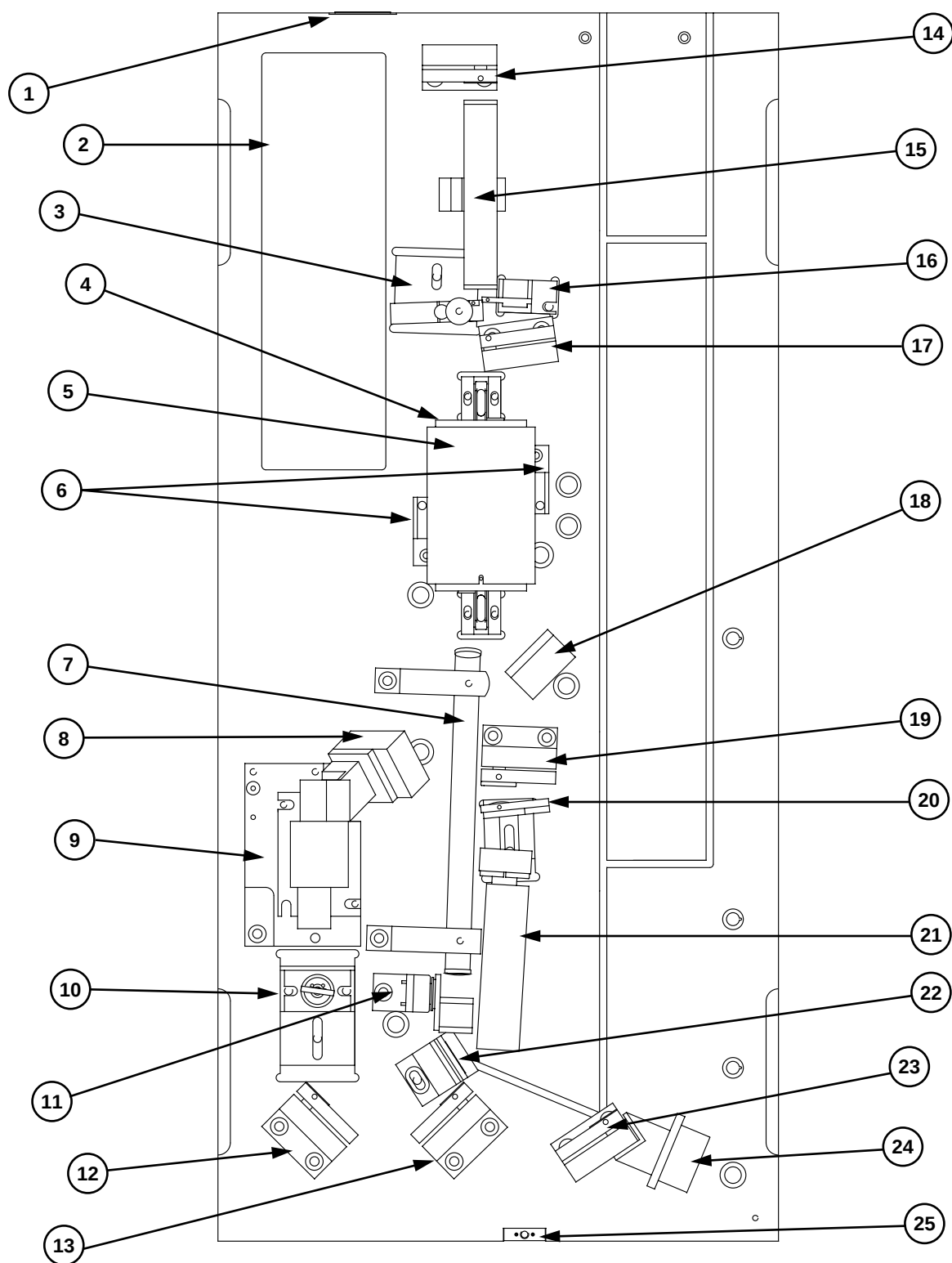


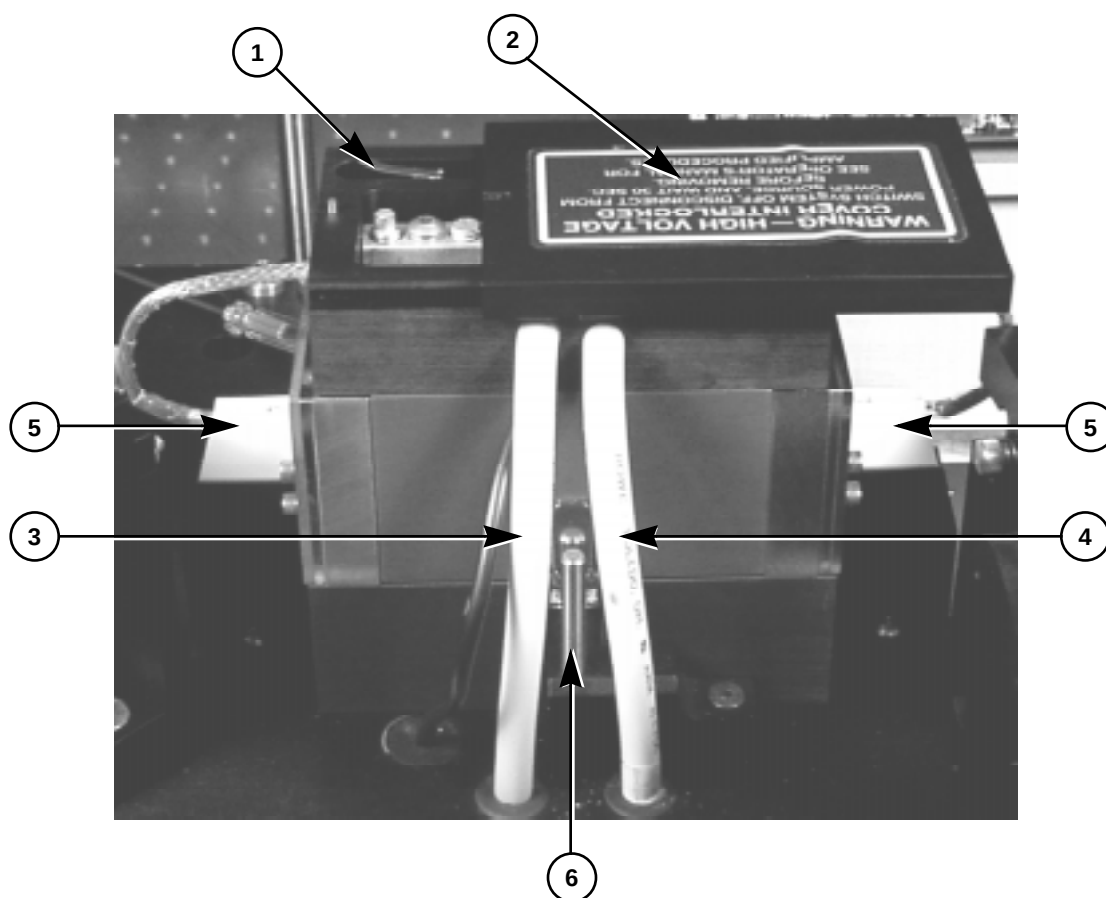
Figure 3-1. Laser Head Controls and Indicator Locations.

Table 3-1. Laser Head Controls and Indicators

CALLOUT	DESIGNATION	REFERENCE FIGURE
1	Laser emission indicator	3-1
2	DPMO, Diode Pumped Master Oscillator	3-1
3	Relay system lens #1	3-4
4	Amplifier assembly, POT	3-2
5	Flash lamp terminal box (above amplifier assembly)	3-2
6	Safety interlock dowel pin	3-2
7	Vacuum spatial filter	3-1
8	Depolarized energy detector	3-3
9	Faraday isolator	3-3
10	Collimating lens	3-1
11	Mechanical shutter	3-1
12	Folding mirror, M1	3-1
13	Reversed folding mirror, M2	3-1
14	Concave mirror, relay system	3-1
15	Relay system vacuum cell	3-1
16	Relay system lens 2	3-4
17	90° Polarization rotation prism	3-4
18	Flashtrack photodetector	3-5
19	Quarter wave plate	3-5
20	SBS lens	3-1
21	SBS cell	3-1
22	Thin film polarizer	3-1
23	Folding mirror, M3	3-6
24	Photo-diode monitoring IR energy	3-6
25	Head cover interlock	3-1



If a back reflection, from an optic external to the laser head, is sent into the laser it can be amplified by residual gain in the POT assembly. After amplification the back reflection, even after being attenuated by the Faraday isolator, may have enough energy to damage the master oscillator permanently. Always be aware of the location of back reflections from optical components when moving them in, or through, the Infinity output beam path.

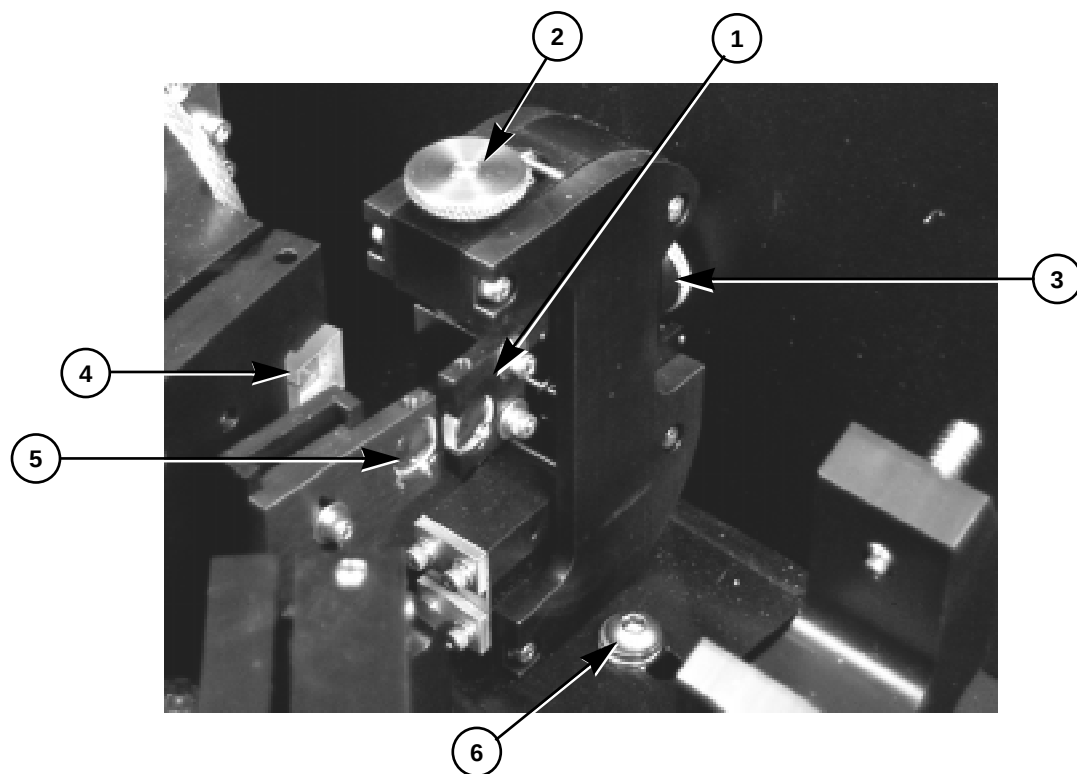


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| 1. Flash lamp terminal box slide cover | 4. Lamp cathode wire |
| 2. Terminal box cover interlock | 5. Teflon lamp end shields (2x) |
| 3. Lamp anode wire | 6. Safety interlock dowel pin (2x) |

Figure 3-2. Infinity POT Assembly.

Table 3-2. Infinity POT Assembly Components

CONTROL	FUNCTION
Terminal box cover interlock	One of three interlocks on the Infinity POT assembly. Opening the terminal box cover with the laser on will result in the laser shutting down and an interlock fault being displayed by the Infinity computer system.
Anode and Cathode wires	Complete the current path, through the lamp, for the capacitor bank discharge to B ⁻ (ground). When verifying or setting the lamp pulse width the cathode wire (closest to relay lens #1) should be used. See Chapter Six, Maintenance, for the complete system timing procedure.
Safety interlock dowel pin	There is an interlock dowel pin on each side of the Infinity POT assembly. Their purpose is to prevent the POT clips from opening during normal operation of the laser system. If the laser is moved these pins should be removed from the head and placed in the "Infinity Maintenance Kit". If a dowel pin is removed while the laser system is running, the system will shut down and an interlock fault will be displayed by the Infinity computer system.

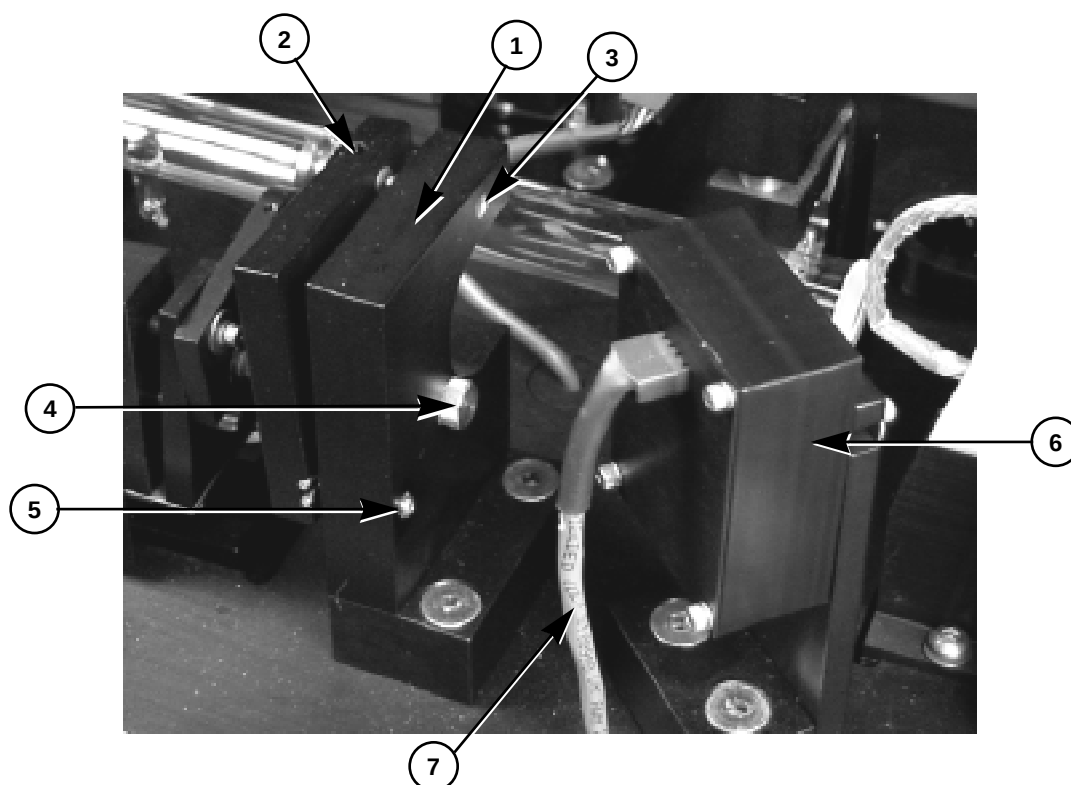


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| 1. Relay lens #1 assembly | 4. 90 degree rotator assembly |
| 2. Relay lens #1 vertical control | 5. Relay lens #2 assembly |
| 3. Relay lens #1 horizontal control | 6. Relay lens #1 translation set screw |

Figure 3-3. Relay Lens #1, #2, and 90 Degree Rotator.

Table 3-3. Relay Lens #1, #2, and 90 Degree Rotator Components

CONTROL	FUNCTION
Relay lens #1 assembly	First lens in the Infinity relay imaging system. This lens can be adjusted to minimize the system depolarized energy at high repetition rates. See Chapter Four, Procedure 4, for additional information.
Relay lens #1 vertical and horizontal adjustments	Allows the vertical and horizontal positioning of the relay lens. See Chapter Four, Procedure 4, for additional information.
Relay lens #1 translation set screw	Locks the translational position of relay lens #1. Under normal operating conditions the factory determined position should not be changed.
Depolarized energy detector bracket retaining screw	Removing this screw allows the depolarized energy detector, including the bracket, to be detached from the Faraday isolator. To prevent damage to the electronics the photodetector electronics plug should be unplugged with the system key switch in the “off “position, before the bracket retaining screw is removed.
90 degree rotator assembly	This optical component can be adjusted to minimize the system depolarized energy. See Chapter Four, Procedure 4, for additional information. See Table 3-4, quarterwave plate, for a description of the Infinity fold assembly vertical and horizontal tilt adjustments.

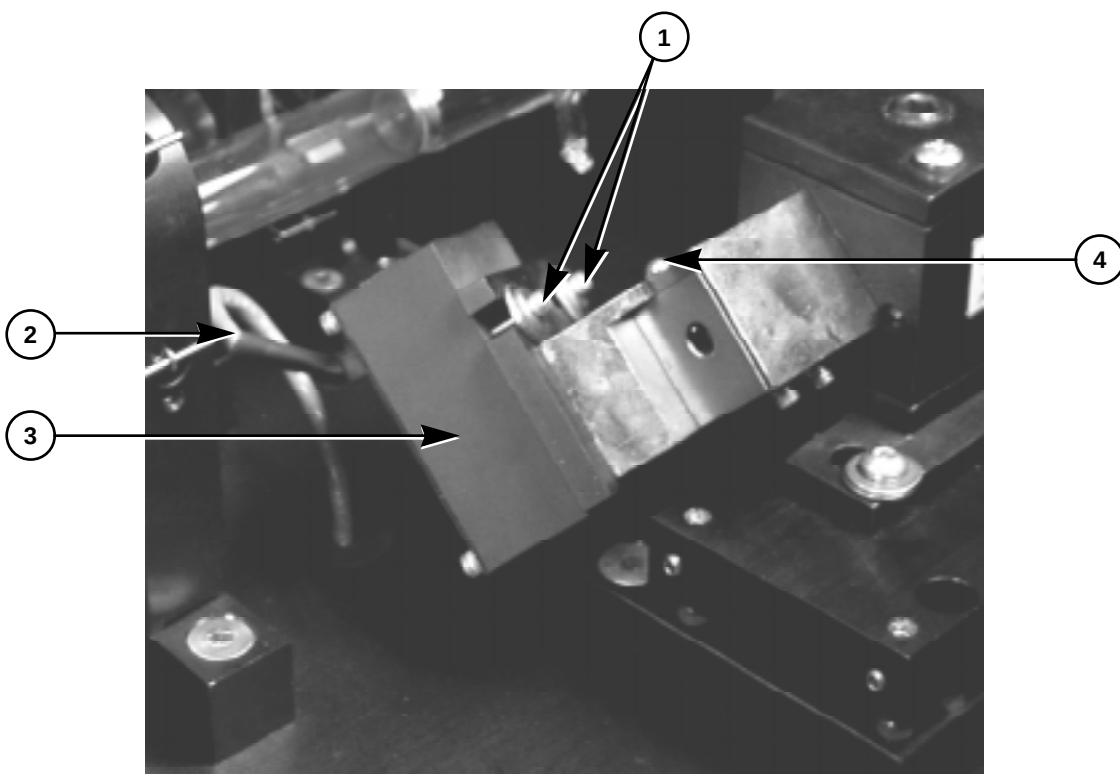


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| 1. Quarter waveplate assembly | 5. Fold assembly horizontal tilt adjustment |
| 2. Quarter waveplate setscrew | 6. Flashtrack photodetector assembly |
| 3. Fold assembly vertical tilt adjustment | 7. Flashtrack photodiode electronics connector |
| 4. Fold assembly tensioning screw | |

Figure 3-4. Quarter Waveplate and Flashtrack Photodetector Assemblies.

Table 3-4. Quarter Waveplate and Flashtrack Photodetector Assembly Components

CONTROL	FUNCTION
Quarter waveplate assembly	Rotates the polarization of the Q-switched DPMO (DPSS) output from horizontal to vertical. This optical component can be adjusted to minimize the system depolarized energy. See Chapter Four, Procedure 4, for additional information.
Quarter waveplate set screw	Secures the quarter waveplate in the fold assembly. Loosening the setscrew will allow the waveplate to be rotated. See Chapter Four, Procedure 4, for additional information.
Fold assembly vertical and horizontal tilt adjustments	Allow for the horizontal and vertical adjustment of the optical component be held in the fold assembly. In this case, that optical component is the quarter waveplate.
Fold assembly tensioning screw	Determines the amount of pressure required to adjust the vertical and horizontal tilt controls of the fold assembly.
Flashtrack photodetector assembly	Provides real time corrections to the lamp flash energy. See Chapter Eight, System Theory, for additional information on this feature.

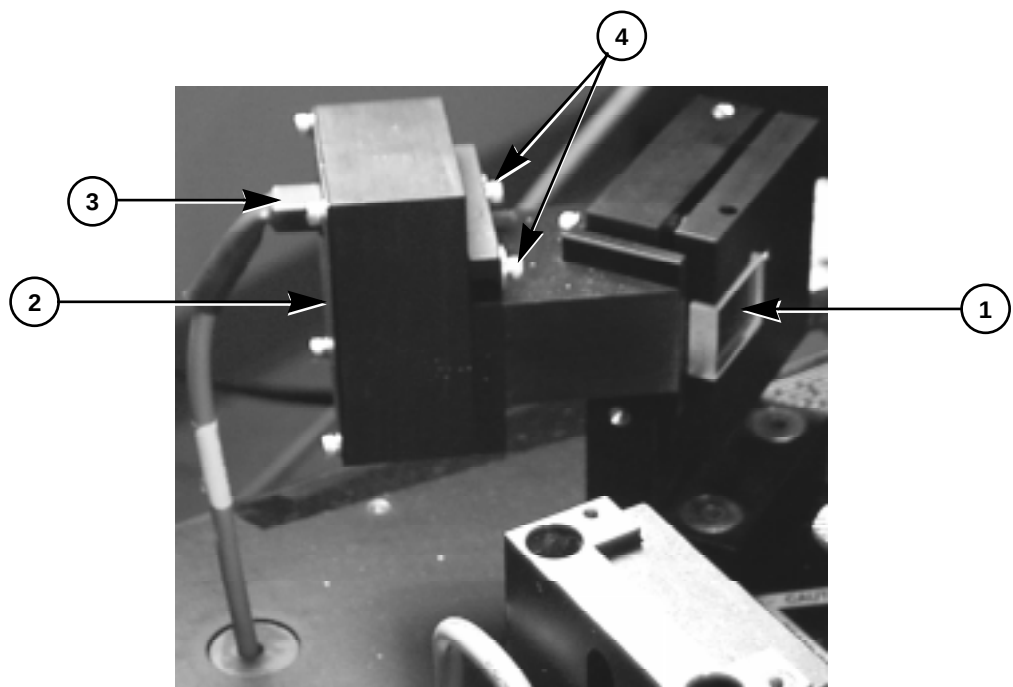


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| 1. Diode housing securing bolts | 3. Photodiode housing |
| 2. Photodiode electronics plug | 4. Depolarized energy detector bracket retaining screw |

Figure 3-5. Depolarized Energy Photodetector.

Table 3-5. Depolarized Energy Detector Components

CONTROL	FUNCTION
Diode housing securing bolts	Loosening these bolts will allow the photodetector signal to be maximized, a step in the calibration procedure for this detector. Under normal operating conditions repositioning of the depolarized energy detector is not required.
Depolarized energy detector bracket retaining screw	Removing this screw allows the depolarized energy detector, including the bracket, to be detached from the Faraday isolator. To prevent damage to the electronics the photodetector electronics plug should be unplugged with the system key switch in the “off” position, before the bracket retaining screw is removed.

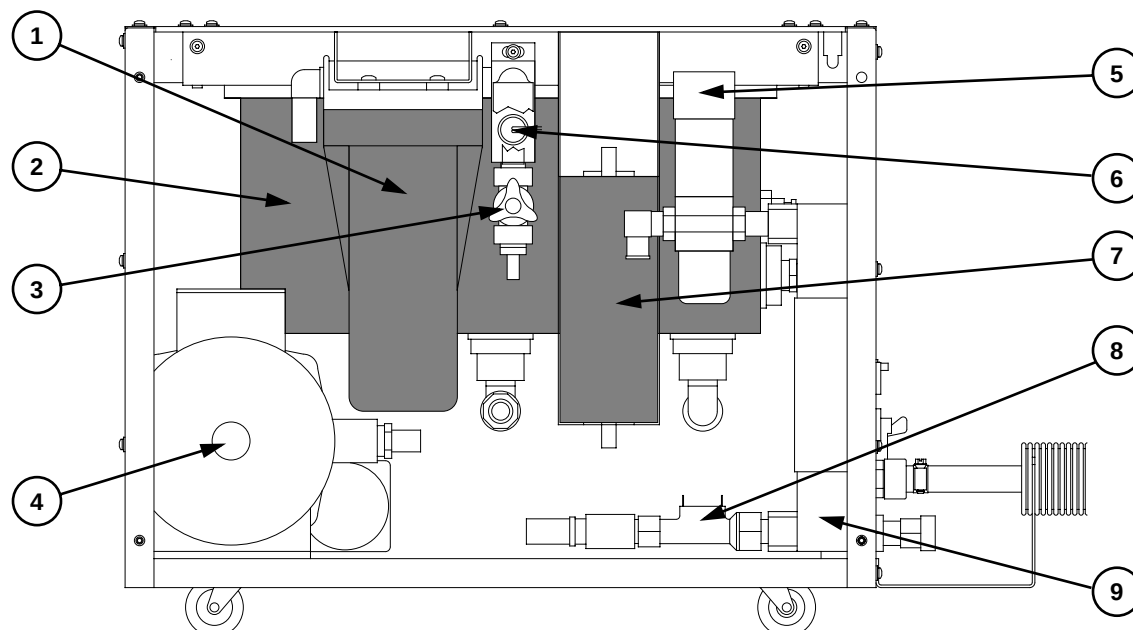


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| 1. IR detector fold assembly | 3. Photodiode electronics plug |
| 2. Photodiode housing | 6. Diode housing securing bolts |

Figure 3-6. IR Photodetector Assembly

Table 3-6. IR Photodetector Assembly Components

CONTROL	FUNCTION
IR fold assembly	Final fold assembly after amplification stage. See Table 3-4, quarter waveplate, for a description of the horizontal and vertical tilt adjustments.
Diode housing securing bolts	Loosening these bolts will allow the photodetector signal to be maximized, a step in the calibration procedure for this detector. Under normal operating conditions repositioning of the depolarized energy detector is not required.

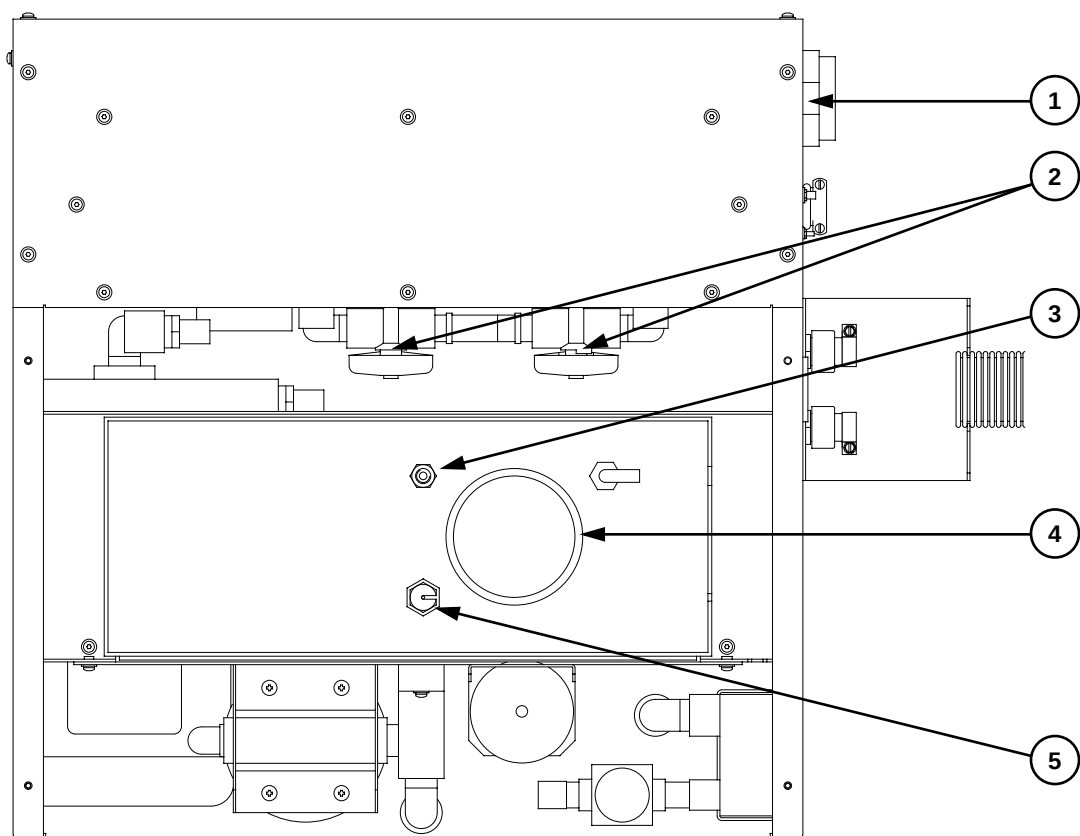


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| 1. Particle filter | 6. Resistivity sensor |
| 2. Water reservoir | 7. Deionization Cartridge |
| 3. Resistivity bypass valve | 8. Water flow switch |
| 4. Water pump | 9. Heat exchanger |
| 5. Temperature regulation valve | |

Figure 3-7. Power Supply Side View, Water Side Components Locations.

Table 3-7. Power Supply Side View, Water Side Components

CONTROL	FUNCTION
Particle filter	Only the particle filter supplied by Coherent should be used in the Infinity laser system. Standard filters, such as those used in ion lasers, can contain manufacturing residues which can greatly affect the performance of the Infinity laser system. See Appendix B, Part Numbers, for the Infinity particle filter order number.
Water reservoir	Volume ~4.5 gallons. Only steam distilled water should be used in the Infinity heat exchanger.
Resistivity bypass valve	Valve controlling the proportion of water going through the deionization cartridge. Turning it clockwise decreases flow which decreases the resistivity.
Temperature regulation valve	Valve controlling flow rate of primary cooling water. Turning it clockwise increases flow, i.e. lowers the temperature of the internal water.



1. Fuse block

2. Booster board water valves

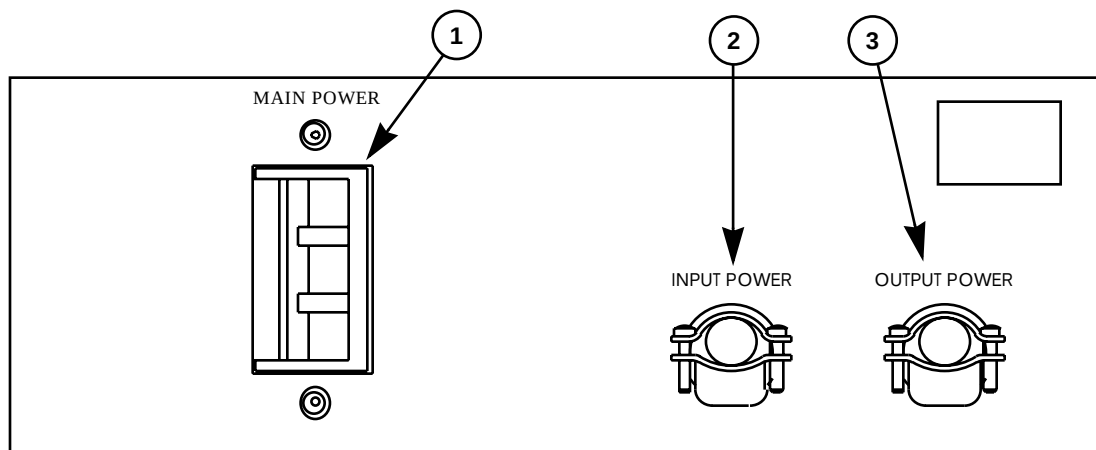
3. Water level sensor
4. Water reservoir fill lid

5. Water temperature sensor

Figure 3-8. Power Supply Top View, Water Side Components Locations.

Table 3-8. Power Supply Top View, Water Side Components

CONTROL	FUNCTION
Fuse block	Contains the fuses for the facility power to the Infinity laser system.
Booster board valves	Under normal operating conditions these values should be in the open position (parallel to the cooling lines). Operating the Infinity laser system with one or both of the valves closed could result in the catastrophic failure of the booster boards.



1. Fuse block
2. 3-phase input line from facility
3. 3-phase output line to Infinity power supply

Figure 3-9. External EMI Filter, Back View.

Table 3-9. External EMI Filter, Back View.

CONTROL	FUNCTION
External EMI filter	To meet CE-MARK requirements an external EMI filter is supplied with systems shipped to Europe. The purpose of this filter is to prevent the Infinity laser system from feeding noise onto the facility power lines. The filter has no effect on the performance of the laser system, and has no internal components which are user serviceable.

